

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS(21CS701HSM)

UNIT – 2

DEMAND AND SUPPLY ANALYSIS

TOPIC- 1 DEMAND MEANING AND DEFINITION

- **What is Demand?**

“ Every want supported by the willingness and ability to buy constitutes demand for a particular product or services”.

- A product or services is said to have demand when 3 conditions are satisfied:
 - ❖ Desire on the part of the buyer to buy
 - ❖ Willingness to pay for it.
 - ❖ Ability to pay the specified price for it. Eg; to buy a car

Definition of demand

- The quantity of goods & services that consumers are willing and able to buy at a given price during a given time period.
- Pointers- a) Effective desire
 - b) always at a given price
 - c) It's a flowing concept.

Nature of Demand

- Demand always implies at given price.
- The uses and the characteristics of a products affect their demand.
- In other words, a product with more number of uses is naturally more in demand than one with single use.
- The nature of demand is better understood with the variations.

TYPES OF DEMAND

1. Individual demand Vs market demand

The main difference between individual demand and market demand is that individual demand is for a single person or household, while market demand is for all consumers in a market:

- Individual demand

The amount of a product a single person or household is willing to buy based on their income, preferences, and expectations.

Eg; milk purchased by an individual,

- Market demand

The total amount of a product all consumers in a market are willing and able to buy.

where as total milk purchased by all the house holds at a given price , particular period of time.

2. Consumer goods Demand Vs Producer goods Demand /Capital goods

- Consumer goods are finished products that consumers buy for their own use, such as food, clothing, electronics, furniture, and automobiles. They are typically purchased from retail outlets or online.

Eg; food ,electronic goods, furniture etc.

- Capital goods are long-lasting assets that businesses use to produce goods and services, such as buildings, machinery, equipment, vehicles, and tools. They are not usually sold to consumers, but are used to create the products that consumers buy.

3.Perishable goods Demand Vs Durable goods demand

- Perishable or non-durable goods refer to the goods that have a single use With less shelf life
Eg; milk, vegetables, fruits.....
- On the other hand, durable goods refer to goods that can be used repeatedly over a considerable period of time. with more shelf life

Eg; furniture ,electronic goods.,clothes...

4. Firm goods demand Vs Industry goods demand (group of related firm)

- Firm demand is the demand for a specific firm's products or services.
Eg; coca-cola
- Industry demand: The total demand for all products in an industry, which includes products from all firms in the industry.

Eg; coca –cola is a small portion of the beverages industry

ie..it includes demand for pepsi, demand for sprite.....etc that all come under the beverages category.

5.Autonomous demand Vs Dependent goods demand

Direct goods demand Vs Indirect goods demand /Derived demand

- Autonomous demand, also known as direct demand, is when the demand for a product is independent of all other goods in the market.

Eg;Constructing a building

- Derived demand: Demand for a product that is directly related to the demand for another product

Eg; Labour , Material , required for construction of the building.

6. Substitutes goods demand Vs Complementary goods demand/Joint commodities

- Substitute goods are products or services that consumers can use in place of one another because they are similar enough to satisfy the same needs.
Eg;Sugar & jaggary , tea & coffee .
- Joint demand: Demand for products or services that are complementary
Book & pen, car & petrol

7.Short run demand Vs Long run demand

- , In economics, short-run demand is the immediate response to price and income changes
Eg; seasonal products , Where labour is increased to meet the current demand
- demand that exists after the market has successfully adjusted to new products & prices
Eg; buy more machinery and employ people to increase the production more.

8. New demand Vs replacement demand

- The expansion of demand for product or service done through advertising ,venturing into new markets....

- the demand for goods & services to replace existing ones due to being outdated or wear & tear.....

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TOPIC-2 FACTORS DETERMINING DEMAND

The demand for a good increases or decreases depending on several factors. This includes the product's price, perceived quality, advertising spend, consumer income, consumer confidence, and changes in taste and fashion.

These factors can be positive or negative, and they can shift the demand curve to the left or right, indicating an increase or decrease in demand.

We have compiled a list of the top ten factors affecting demand for you.

a) Price of the product or service (P)

.. if price increases, demand for the commodity decreases & vice-versa (inverse relationship)

b) Income of the consumer (I)

Rising incomes lead to a rise in the number of goods demanded by consumers. Similarly, a drop in income is accompanied by reduced consumption levels. This relationship between income and demand is not linear in nature. Marginal utility determines the proportion of change in the demand levels.

c) Tastes and preferences: Consumer preferences, which can be shaped by culture, information, and knowledge of products and services (T)

Either increase or decrease

d) Prices of related goods and services (Pr)

Related goods can be substitute goods or complementary goods or joint goods

- **Complementary products** – An increase in the price of one product will cause a decrease in the quantity demanded of a complementary product. Example: Rise in the price of bread will reduce the demand for butter. This arises because the products are complementary in nature.

- **Substitute Product** – An increase in the price of one product will cause an increase in the demand for a substitute product. Example: Rise in price of tea will increase the demand for coffee and decrease the demand for tea.

e) Consumer expectations for a change in price (E_p)

Expectations in an increase in prices of goods will lead to an increase the quantity demanded. & lowering in prices of goods will decrease the quantity demanded.

f) Consumer expectations for a change in income (E_i)

Expectations of a higher income will lead to an increase the quantity demanded. Similarly, expectations of a reduced income will decrease the quantity demanded.

g) Size of population (S_p)

The number of buyers has a major effect on the total or net demand. As the number increases, the demand rises. Furthermore, this is true irrespective of changes in the price of commodities.

Directly proportionate

h) Distribution of consumers over different regions (D_c).

Distribution of consumers based on their incomes

live in a particular region, their cultures ,traditions....etc. will have an impact on the demand for the goods & services.

i) Advertising efforts (A).

Advertising can educate consumers about a product or service, and reshape their preferences. This can lead to increased demand for the advertised product.

always (+)'ve

j) Any other factors capable of affecting or influencing the demand (O).

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Topic -3

Law of Demand

Definition

Demand is a vital economic concept that works both at the market level and personal level. It also includes several concepts like law of demand, factors affecting it and eventually the impact of it on the economy at large.

The law of demand in economics explains that when other factors remain constant, the quantity demand and price of any product or service show an inverse equation. It also means that whenever the value of a specific product increases, demand for the same declines.

“Other things remaining the same or equal the lower the price at which a thing is offered, the more a man will be prepared to buy it”

In this definition, the phrase, “other things remaining the same” is very important qualifying phrase of law. It is because the demand is not only a function of price alone. Demand depends upon many things like population, income, taste, habit and distribution of income in the society etc

The quantity demanded varies inversely with price.

It is usually represented as $QD = f \{P\}$

Where QD is quantity demanded

F is Function of

P is the Price

Which in simpler terms mean the quantity demanded for any product or service would directly depend upon the price.

Assumptions to the Law of Demand

1.The income of the consumer: One of the most important assumptions to this law is that the incomes of the consumers do not change, that is it will not increase. As the income of the consumer increases there is a tendency to buy more so a very important assumption for the theory/law to prove good is the income of the consumer won't change.

2.Consumers tastes and preferences remain same. If the tastes and preferences of the consumers change, then there may be a change in the quantity demanded irrespective of the increase in price and hence it is assumed that the tastes and preferences of the consumers remain same or constant.

3.Prices of related goods remain unchanged: The Prices of related goods here means both substitutes and complimentary goods. Substitute Goods and Complementary Goods are two economic concepts describing the relationship between two or more different products in terms of their demand and consumption patterns. Substitute goods are the goods that can be used in place of one another; however, Complementary goods are the goods that can be used together.

Eg:The demand for product X is determined by the prices of its related products.

If there is an increase in the price of substitute, the demand for product X will go up and vice versa.

4.Consumer expectation do not change: People do not expect the future price of the commodity to change

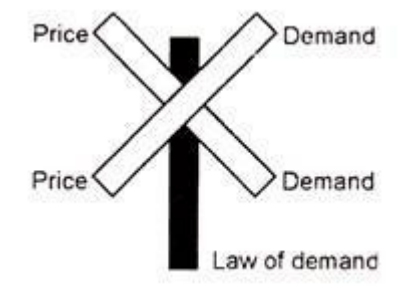
5.population should be constant:The population of the country should be constant. This necessitates that the size of population as well as its age. Structure and sex ratio should remain the same throughout the operation of the law. Otherwise, if population changes, there will be additional buyers in the market, so the total market demand may not contract with a rise in price

6.No change in government policy: The level of taxation and fiscal policy of the government should remain the same throughout the operation of the law. Otherwise, the changes in income-tax for instance, may cause changes in consumer's income or commodity taxes like sales tax or excise duties may lead to distortions in consumer's preferences. Therefore, the validity of the Law of Demand or the inference about inverse relationships between price and demand depends on the existence of these conditions or assumptions.

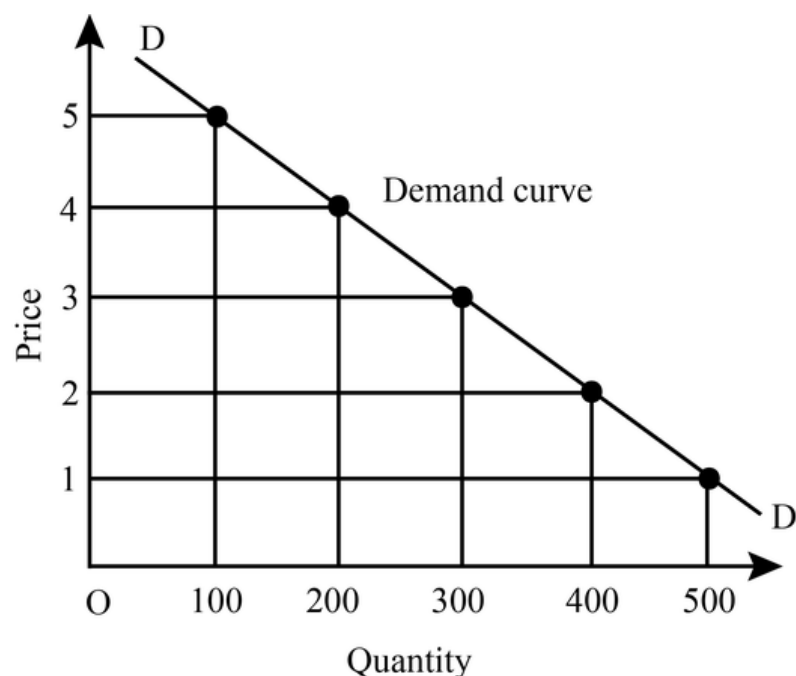
Explanation of Law of Demand/Operation of Law of Demand

It is the view of economists that the Law of Demand is based on Diminishing Marginal Utility. This law simply states that as the price of a commodity increases demand reduces and vice-versa. Consumer wants to pay the price of a commodity up to the extent of marginal utility. Therefore, an increase in the quantity of a commodity, its utility diminishes, slowly and consumer likes to pay less price of that commodity.

In other-words, it can be said that if the price of a commodity reduces additional unit of goods can be purchased. Eminent economist like Prof. Marshall has compared this law with a game of “See-Saw” in which he has stated that when one end of a plank of wood goes up the other end goes down which can be seen from the diagram given below:



Demand Schedule	
Px	Dx
5	100
4	200
3	300
2	400
1	500



From the above graph, the following points to be noted

- Law of demand may be explained with the help of the following demand schedule and demand curve:

- If the price of the product is Rs 3 (P) the quantity demanded is 300 units (Q)
- If the price of the product is increased to Rs 5 (P1) the quantity demanded is reduced to 100 units (Q1)
- If the price of the product is reduced to Rs 1 (P2) the quantity demanded is increased to 500 units (Q2).
- Demand curve will be sloping downwards from left to right.

Demand Curve will always slope from left to right.

Exceptions to law of demand:

1.In case of Emergencies:The law of demand does not apply in case of an emergency. This may be caused by war, famine political or social Upheaval. During these periods consumers behave in an abnormal way. They accentuate scarcities and induce further price rises by making more purchases even at higher prices. This can also be caused by emergency of wants also.

2.Expectations of Future prices: When we expect that the price of any commodity will increase in the near future with the prices that are rising now. In that case we shall buy more at increasing prices. If we anticipate that the price will fall in the near future even with the fall in price at present we shall buy less. This is contrary to the Law of Demand.

3.Prestigious goods/Veblen Goods or Luxurious Goods: Products such as jewels, diamonds and so on, confer distinction on the part of user. In such a case consumer tends to buy even though there is a increase in price. Such goods are called Veblen goods.

4.Giffen Paradox/Giffen Goods:Sir Robert Giffen has observed that there is a strange relationship between price and quantity demanded and thus, consumers buy less of a commodity at a lower price and more of a commodity at higher price. He has cited the example of low paid British wage earners. During the early period of nineteenth century, a rise in the price of bread caused the wage earners to buy the same amount of bread as before. This tendency is generally known as “Giffen Paradox.”

5.Ignorance of Price changes: Consumer's ignorance also affects the quantity demanded. He purchases more of the commodity at a higher price. This is especially so when the consumer is haunted by the phobia that a higher priced commodity is better in quality and low priced commodity is inferior

An Exceptional Demand Curve

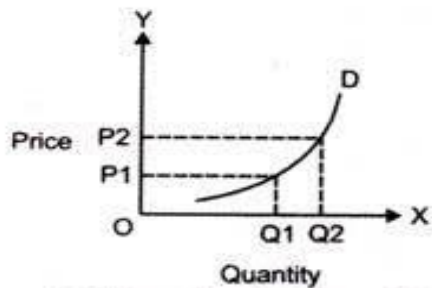


Figure-10: Exceptional Demand Curve

In case of exceptions the demand curve slopes upwards.

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TOPIC-4 CHANGE IN DEMAND

Change in demand can be :

- 1) Increase or decrease in demand.(price is constant & other determinants are varying)
- 2) Expansion and Contraction in demand (price will be varying & other determinants kept constant)

1) Increase or decrease in demand.

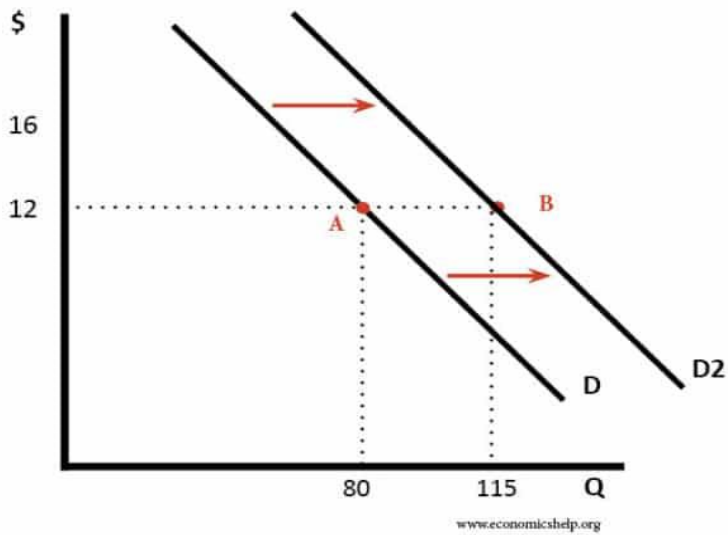
When there is a change in the quantity demanded of a particular commodity, at each possible price, due to a change in one or more other factors, the demand curve **shifts**. The important aspect to remember is that other factors like the consumer's income and tastes along with the prices of other goods, etc., which were expected to remain constant, changed.

This is the Shift of the Demand Curve. The demand curve can shift either to the left or the right, depending on the factors affecting it.

Increase in Demand

❖ Increase in Demand:

- **If the consumer are willing and able to buy more no. of products at the same price, the result will be an increase in demand.**
- **The demand curve will shift to the right side.**



Decrease in Demand

- A decrease in demand occurs when buyers are ready to buy less of a product at the same price because of factors like fall income , rise in price of complementary goods and so on.
- Decrease in demand will shift the demand curve to the left.
- **Increase or decrease in the demand involves a shift in the demand curve.**



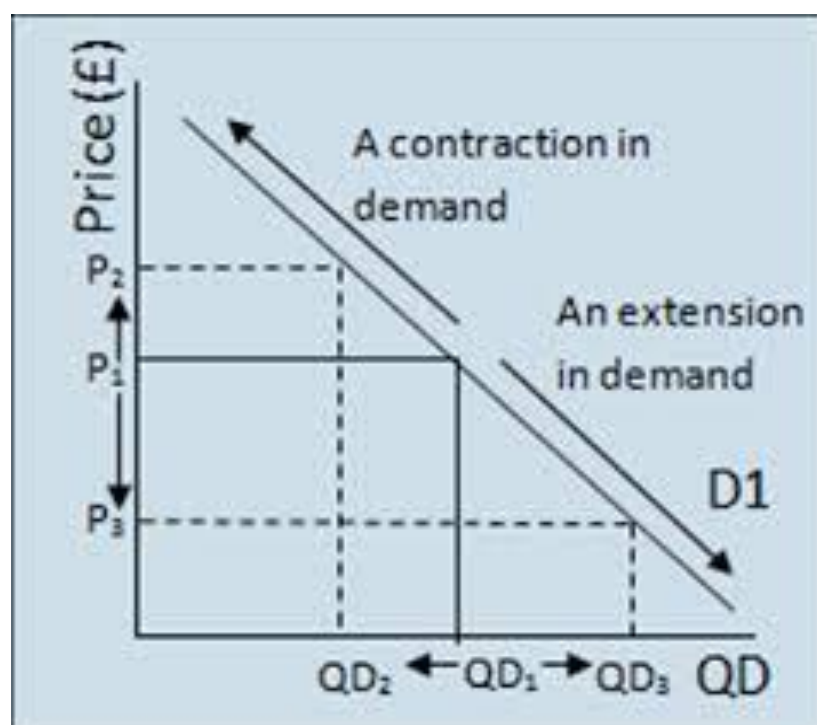
Extension and contraction in Demand

- Extension and contraction in a demand curve will be observed due to change in the price of a commodity.
- A **Extension** is the downward movement along a demand curve.

Which indicates that a higher quantity is demanded for a given fall in the price of the good.

- A **Contraction** is the upward movement along a demand curve.
- Which indicates that a lower quantity is demanded for a given increase in the price of the good.
- There is **no shift** in the demand curve.

Graph representing Extension and Contraction in Demand



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TOPIC-5 ELASTICITY OF DEMAND

ELASTICITY OF DEMAND

Elasticity of demand explains the relationship between a change in price and consequent change in amount demanded. “Marshall” introduced the concept of elasticity of demand. Elasticity of demand shows the extent of change in quantity demanded to a change in price & other determinants.

Measurement of Elasticity

The Elasticity is measured in the following ways

a)

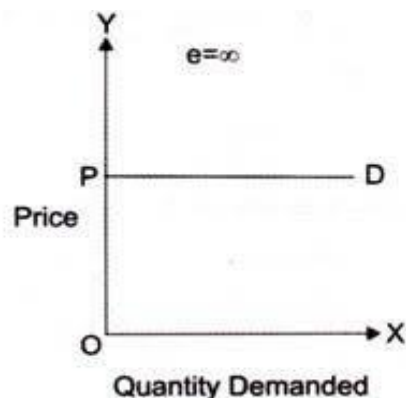


Figure-2: Perfectly Elastic Demand

Perfectly elastic demand

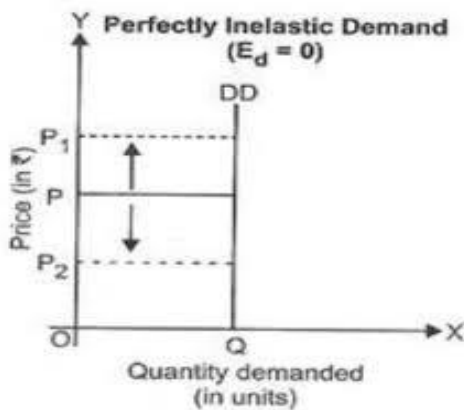
- When small change in price leads to an infinitely large change in quantity demand, it is called perfectly or infinitely elastic demand.

- In this case $E_d = \infty$

- The demand curve DD1 is horizontal straight line.

It shows the at “OP” price any amount is demand and if price increases, the consumer will not purchase the commodity.

b) perfectly inelastic



- In this case, even a large change in price fails to bring about a change in quantity demanded.
- When price increases from 'OP' to 'O1P', the quantity demanded remains the same.
- In other words the response of demand to a change in Price is nil. In this case ' $E_d=0$ '.

c) Relatively Elastic Demand

- Demand changes more than proportionately to a change in price. i.e. a small change in price leads to a very big change in the quantity demanded.
- In this case $E_d > 1$. This demand curve will be flatter.

When price falls from 'OP1' to 'OP2', quantity demanded increase from "OQ1" to "OQ2" which is larger than the change in price

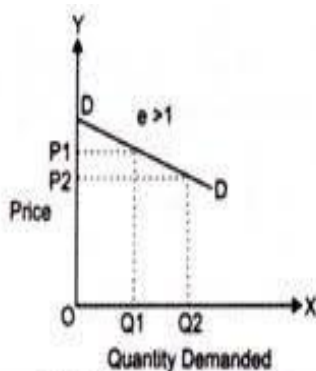


Figure-4: Relatively Elastic Demand

d). Relatively in-elastic demand.

- Quantity demanded changes less than proportional to a change in price.
- A large change in price leads to small change in the quantity demanded.
- Here $E_d < 1$. Demand curve will be steeper.
- When price falls from “OP1” to ‘OP2’ quantity demanded increases from OQ1 to OQ2, which is smaller than the change in price.

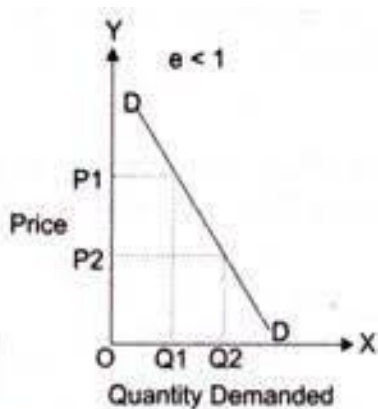


Figure-5: Relatively Inelastic Demand

e) Unit elasticity of demand:

- The change in demand is exactly equal to the change in price. When both are equal $E_d = 1$ and elasticity is said to be unitary.
- When price falls from ‘OP1’ to ‘OP2’ quantity demanded increases, from ‘OQ1’ to ‘OQ2’.
- Thus a change in price has resulted in an equal change in quantity demanded so price elasticity of demand is equal to unity
- .

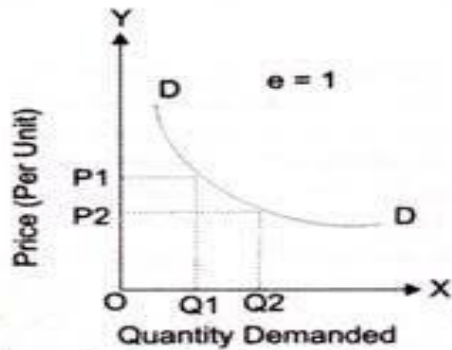


Figure-6: Unitary Elastic Demand

Importance of Elasticity of Demand:

The concept of elasticity of demand is of much practical importance.

- **Price fixation:** Each seller under monopoly and imperfect competition has to take into account elasticity of demand while fixing the price for his product. If the demand for the product is inelastic, he can fix a higher price.

Production: Producers generally decide their production level on the basis of demand for the product. Hence elasticity of demand helps the producers to take correct decision regarding the level of output to be produced.

- **Distribution:** Elasticity of demand also helps in the determination of rewards for factors of production. For example, if the demand for labour is inelastic, trade unions will be successful in raising wages. It is applicable to other factors of production.
- **International Trade:** Elasticity of demand helps in finding out the terms of trade between two countries. Terms of trade refers to the rate at which domestic commodity is exchanged for foreign commodities. Terms of trade depends upon the elasticity of demand of the two countries for each other goods.
- **Public Finance:** Elasticity of demand helps the government in formulating tax policies. For example, for imposing tax on a commodity, the Finance Minister has to take into account the elasticity of demand.
- **Nationalization:** The concept of elasticity of demand enables the government to decide about nationalization of industries.

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TOPIC -6 TYPES OF ELASTICITY OF DEMAND

ELASTICITY OF DEMAND

Elasticity of demand explains the relationship between a change in price and other determinants with consequent change in amount demanded.

“Marshall” introduced the concept of elasticity of demand.

Elasticity of demand shows the extent of change in quantity demanded to a change in price.

In the words of “Marshall”, “The elasticity of demand in a market is great or small according as the amount demanded increases much or little for a given fall in the price and diminishes much or little for a given rise in Price”

Elastic demand: A small change in price may lead to a great change in quantity demanded. In this case, demand is elastic.

In-elastic demand: If a big change in price is followed by a small change in demanded then the demand is “inelastic”.

Types of Elasticity of Demand:

There are three types of elasticity of demand:

1. Price elasticity of demand
2. Income elasticity of demand
3. Cross elasticity of demand
4. promotional Elasticity of demand

1. Price elasticity of demand:

Marshall was the first economist to define price elasticity of demand. Price elasticity of demand measures changes in quantity

demand to a change in Price. It is the ratio of percentage change in quantity demanded to a percentage change in price.

Edp = Proportionate change in Qty demanded

Proportionate change in the price

= Q2-Q1 / Q1

P2-P1 / P

There are five cases of price elasticity of demand

- a) Perfectly elastic
- b) Perfectly inelastic
- c) Relatively elastic
- d) Relatively inelastic
- e) Unitary elastic

2. Income elasticity of demand:

Income elasticity of demand shows the change in quantity demanded as a result of a change in income. Income elasticity of demand may be stated in the form of a formula.

Proportionate change in the quantity demanded of a product

Edi = $\frac{\text{Proportionate change in the quantity demanded of a product}}{\text{Proportionate change in the income}}$

= $\frac{Q_2 - Q_1}{Q_1}$

$\frac{I_2 - I_1}{I_1}$

Income elasticity helps in knowing whether the commodity is a:

1. Superior good ($E_d > 1$)
2. Normal good or a ($E_d \leq 1$)
3. Inferior good ($E_d < 0$)

When income increases quantity demanded also increases but less than proportionately. In this case $E_d < 1$. An increase in income from OY to OY_1 , brings what an increase in quantity demanded from OQ to OQ_1 , But the increase in quantity demanded is smaller than the increase in income. Hence, income elasticity of demand is less than one.

3. Cross elasticity of Demand:

A change in the price of one commodity leads to a change in the quantity demanded of another commodity. This is called a cross elasticity of demand. The formula for cross elasticity of demand is:

$E_{dxy} = \frac{\text{Proportionate change in the qty demanded of product 'x'}}{\text{Proportionate change in the price of product 'y'}}$

$$E_{dxy} = \frac{Q2x - Q1x}{Q1x}$$

$$\frac{P2y - P1y}{P1y}$$

Q1x--- Initial qty demanded of product 'x'

Q2x---New qty demanded of product 'x'

P1y –Initial price of product 'y'

P2y –changed price of product 'y'

A .In case of substitutes, cross elasticity of demand is positive. Eg: Coffee and Tea

When the price of coffee increases, Quantity demanded of tea increases. Both are substitutes. Price of Coffee

b. In case of compliments, cross elasticity is negative. If increase in the price of one commodity leads to a decrease in the quantity demanded of another and vice versa When price of car goes up from OP to OP!, the quantity demanded of petrol decreases from OQ to OQ!. The cross-demanded curve has negative slope.

c. In case of unrelated commodities, cross elasticity of demanded is zero. A change in the price of one commodity will not affect the quantity demanded of another.

Quantity demanded of commodity “b” remains unchanged due to a change in the price of ‘A’, as both are unrelated goods.

4 . Advertising elasticity of demand

It refers to increase in the sale revenue because of changes in the advertising expenditure. In other words there is a direct relationship between the amount of money spent on advertising and its impact on sales. It is always positive

Advertising elasticity greater than unity:

$$E_{dA} = \frac{Q2 - Q1}{Q1}$$

$$\frac{A2 - A1}{A1}$$

Q1 & Q2—Initial & changed qty demanded.

A1 & A2--- Initial & changed Advertising expenses.

Factors effecting Elasticity of Demand

Elasticity of demand depends on many factors.

1. Nature of commodity: Elasticity or in-elasticity of demand depends on the nature of the commodity i.e. whether a commodity is a necessity, comfort or luxury, normally; the demand for Necessaries like salt, rice etc is inelastic. On the other band, the demand for comforts and luxuries is elastic.

2. Availability of substitutes: Elasticity of demand depends on availability or non-availability of substitutes. In case of commodities, which have substitutes, demand is elastic, but in case of commodities, which have no substitutes, demand is in elastic.

3. Variety of uses: If a commodity can be used for several purposes, than it will have elastic demand. i.e. electricity. On the other hand, demanded is inelastic for commodities, which can be put to only one use.

4. Amount of money spent: Elasticity of demand depends on the amount of money spent on the commodity. If the consumer spends a smaller for example a consumer spends a little amount on salt and matchboxes. Even when price of salt or matchbox goes up, demanded will not fall.

5. Postponement of demand: If the consumption of a commodity can be postponed, than it will have elastic demand. On the contrary, if the demand for a commodity cannot be postponed, than demand is in elastic. The demand for rice or medicine cannot be postponed, while the demand for Cycle or umbrella can be postponed.

6. Time: Elasticity of demand varies with time. Generally, demand is inelastic during short period and elastic during the long period. Demand is inelastic during short period because the consumers do not have enough time to know about the change is price. Even if they are aware of the price change, they may not immediately switch over to a new commodity, as they are accustomed to the old commodity.

7. Range of Prices: Range of prices exerts an important influence on elasticity of demand. At a very high price, demand is inelastic because a slight fall in price will not induce the people buy more. Similarly at a low price also demand is inelastic. This is because at a low price all those who want to buy the commodity

would have bought it and a further fall in price will not increase the demand.
Therefore, elasticity is low at very high and very low prices.

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TOPIC- 7 DEMAND FORECASTING AND METHODS OF DEMAND FORECASTING

DEMAND FORECASTING

INTRODUCTION:

The information about the future is essential for both new firms and those planning to expand the scale of their production. Demand forecasting refers to an estimate of future demand for the product. It is an ‘objective assessment of the future course of demand’. In recent times, forecasting plays an important role in business decision-making.

TYPES OF DEMAND FORECASTING:

Based on the time span and planning requirements of business firms, demand forecasting can be classified in to

1. Level of forecasting—
 - a) Firm level
 - b) Industry level
 - c) Economy level

2. Time period involved---
 - a) Short-term demand forecasting
 - b) Long - term forecasting

a) Short-term demand forecasting:

- Short-term demand forecasting is limited to short periods, usually for one year.
- It relates to policies regarding sales, purchase, price and finances. It refers to existing production capacity of the firm.

- Short-term forecasting is essential for formulating a suitable price policy.
- If the business people expect a rise in the prices of raw materials or shortages, they may buy early.

b) Long - term forecasting:

- In long-term forecasting, the businessmen should know about the long-term demand for the product.
- Planning of a new plant or expansion of an existing unit depends on long-term demand.
- Similarly a multi product firm must take into account the demand for different items.
- When forecasts are made covering long periods, the probability of error is high.
- It is very difficult to forecast the production, the trend of prices and the nature of competition. Hence quality and competent forecasts are essential.

3. Nature of the goods /products----

a) Consumer goods forecasting.
b) Producer goods forecasting.

METHODS OF FORECASTING:

Several methods are employed for forecasting demand. All these methods can be grouped under survey method and statistical method. Survey methods and statistical methods are further subdivided into different categories.

1. Survey Method:

a) Census method

- Also known as complete enumeration this method involves collecting information from every individual in a population.
- Census data is detailed and can be used for policy-making, planning, and more.
- However, it can be expensive and time-consuming.

- **b) Sample survey**

- This method involves randomly selecting a sample from a population and collecting information from them.
- The data is then used to make inferences about the entire population.
- Sample surveys are less expensive and time-consuming than census surveys, and they can enable more detailed questions to be asked.
- However, they are less accurate and reliable than census surveys, and they can be susceptible to personal biases.

c) Opinion survey method:

- This method is also known as sales-force composite method (or) collective opinion method.
- Under this method, the company asks its salesman to submit estimate of future sales in their respective territories.
- Since the forecasts of the salesmen are biased due to their optimistic or pessimistic attitude ignorance about economic developments etc. these estimates are consolidated, reviewed and adjusted by the top executives. In case of wide differences, an average is struck to make the forecasts realistic.
-

d) Expert opinion method:

Apart from salesmen and consumers, distributors or outside experts may also be used for forecasting. In the United States of America, the automobile companies get sales estimates directly from their dealers. Firms in advanced countries make use of outside experts for estimating future demand. Various public and private agencies all periodic forecasts of short or long term business conditions.

e) Consumers interview method:

In this method the consumers are contacted personally to know about their plans and preference regarding the consumption of the product.

A list of all potential buyers would be drawn and each buyer will be approached and asked how much he plans to buy the listed product in future. He would be asked the proportion in which he intends to buy. This method seems to be the most ideal method for forecasting demand.

2. **Statistical Methods:**

Statistical method is used for long run forecasting. In this method, statistical and mathematical techniques are used to forecast demand. This method relies on past data.

a. **Time series analysis or trend projection methods:**

- A well-established firm would have accumulated data. These data are analyzed to determine the nature of existing trend.
- Then, this trend is projected into the future and the results are used as the basis for forecast.
- This is called as time series analysis. This data can be presented either in a tabular form or a graph.
- In the time series past data of sales are used to forecast future.

b) **Exponential smoothing**

- Exponential smoothing is a weighted moving average technique which is especially effective when frequent re-forecasting is required, and when the forecasts must be achieved quickly.
- It is a short-term forecasting technique that is frequently used in the production and inventory environment, where only the next period's value is required to be forecast.
- Because only three numbers are required to perform exponential smoothing, this technique is simple to update.
- The data required are the historic observation, the latest data observation and the smoothing coefficient, or constant.
- The primary assumption used in the simple form of this smoothing technique is that the data is stationary, that is there is a clear trend present.

Advanced exponential smoothing techniques are required if a trend or cycle is present in the data.

- The algebraic formula for simple exponential smoothing is:

$$F_t = X_t + (1 - \alpha)F_{t-1}$$

Should be $F_t = X_t \cdot \alpha + (1 - \alpha)F_{t-1}$ where F_{t-1} = the previous forecast, X_t = the current observation and α = the smoothing coefficient.

c) Trend Projection Method of Demand Forecasting

- The trend projection method of demand forecasting is a statistical technique used to forecast sales based on past trends.
- It assumes that the demand will continue in the same direction as the past trend, with the same or a similar rate of growth or decline.
- This involves fitting a trend line to historical data and extrapolating the line into the future to estimate the trends

d) Barometric Technique:

- Simple trend projections are not capable of forecasting turning points.
- Under Barometric method, present events are used to predict the directions of change in future.
- This is done with the help of economics and statistical indicators. Those are
 - Construction Contracts awarded for building materials
 - (2) Personal income (3) Agricultural Income. (4) Employment (5) Gross national income (6) Industrial Production (7) Bank Deposits etc.
-

e) Regression and correlation method:

- This method establishes a relationship between the dependent variable and the independent variables.

- In our case, the quantity demanded is the dependent variable and income, the price of goods, the price of related goods, the price of substitute goods, etc. are independent variables.
- The regression equation is derived assuming the relationship to be linear.

- Regression Equation:

$Y = a + bX$. Where Y is the forecasted demand for a product or service.

f) Econometric Method of Demand Forecasting

- The econometric method of demand forecasting is a statistical approach to predicting future sales for a product or service based on past sales data and relevant economic and market factors.
- It uses regression analysis and other statistical tools to establish relationships between demand and independent variables such as economic indicators, market trends, and other relevant variables.
- The model is then used to predict future trends based on expected changes in the independent variables.
- The econometric method is considered highly reliable and accurate, especially for products or services with a long sales history.
- However, it requires a large amount of data and a good understanding of statistical techniques to implement effectively.

OTHER METHODS

a) Delphi Method of Demand Forecasting

- The Delphi method is a group consensus-based approach for demand forecasting. It involves collecting predictions and opinions from a panel of experts and then using an iterative process to reach a consensus forecast.
- It involves identifying a panel of experts, collecting initial projections from each expert, compiling, and aggregating the initial estimates, providing expert feedback on the overall results, collecting updated forecasts from the experts, and creating a consensus demand forecast.
- The Delphi method is helpful for complex forecasting problems where a single expert's forecast may need to be more reliable.

b) Market Test Method of Demand Forecasting

- The Market Test method of demand forecasting involves conducting a limited-scale launch of a new product or service in a specific geographic area or customer segment to gather info.
- The data collected from the market test is then used to forecast regarding the product or service in a larger-scale launch.
- This is advantageous when launching a new product or entering a new market.
- It provides real-world information on customer demand, which can be difficult to estimate using other forecasting techniques.
- However, it can be expensive and time-consuming, and results may not represent trends for other regions.
- This is often used with other forecasting methods to increase accuracy.

c) Controlled experiment

The idea behind a controlled experiment is to create a specific environment, a laboratory, in which the relevant variables are manipulated in order to take measurements of particular parameters, with the aim to discover the relationship, if any, between these variables

d).Judgemental approach

when there is a complete lack of historical data, or when a new product is being launched, or when a new competitor enters the market, or during completely new and unique market conditions. judgmental forecasting is the only option.

STEPS IN DEMAND FORECASTING

Specifying the objective

Determining the time perspective

Selecting the method for forecasting

Collecting and adjusting data

Interpreting the outcomes

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS(21CS701HSM)

UNIT-2

TOPIC-8 SUPPLY ANALYSIS

SUPPLY

- **What is Law of Supply?**
- Economists have studied the behaviour of both buyers and sellers. They have discovered the law of supply as a result of their findings. *The **law of supply** describes the relationship between price and amount supplied when all other variables remain constant (*ceteris paribus*).*

Here are some determinants of the supply curve.

1.Price of the commodity.(P)

2.Production cost: (C)

Since most private companies' goal is profit maximization. Higher production cost will lower profit, thus hinder supply. Factors affecting production cost are: input prices, wage rate, government regulation and taxes, etc.

3.Technology: (T)

Technological improvements help reduce production cost and increase profit, thus stimulate higher supply.

4. Government subsidies and taxes. (G)

5. Expectation for future prices: (E)

If producers expect future price to be higher, they will try to hold on to their inventories and offer the products to the buyers in the future, thus they can capture the higher price.

6. Number of sellers/Firms (N)

More sellers in the market increase the market supply.

Supply Function

- Supply function is the mathematical expression of law of supply. In other words, supply function quantifies the relationship between quantity supplied and price of a product, while keeping the other factors at constant. The law of supply expresses the nature of relationship between quantity supplied and price of a product, while the supply function measures that relationship.

•

$$Q_s = f(P, C, T, G, E, N)$$

Where,

Q_s = Supply, P = Price of the good supplied

C = Cost of goods & services

T = Technology, G = Govt. rates & taxes

E = Expected future price

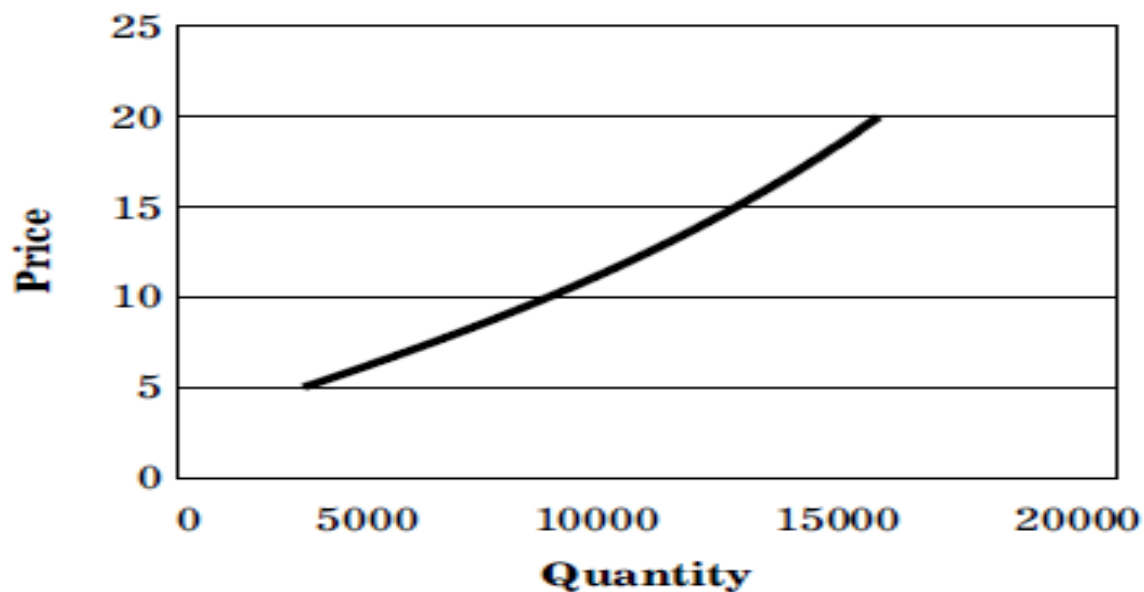
N = other firms selling the same commodity

Definition of 'Law Of Supply'

Definition: Law of supply states that other factors remaining constant, price and quantity supplied of a good are directly related to each other. In other words, when the price paid by buyers for a good rises, then suppliers increase the supply of that good in the market.

Description: Law of supply depicts the producer behavior at the time of changes in the prices of goods and services. When the price of a good rises, the supplier increases the supply in order to earn a profit because of higher prices.

Price of the Product (₹ per Kg)	Quantity Supplied of Commodity A (Kg per Week)
5	3,000
10	8,000
15	12,000
20	15,000



The **supply curve** is a graphical representation of the relationship between the price of a good or service and the quantity supplied for a given period of time. In a typical representation, the price will appear on the left vertical axis, the quantity supplied on the horizontal axis.

The above diagram shows the supply curve that is upward sloping (positive relation between the price and the quantity supplied). When the price of the good was at **P1 (Rs.5)**, suppliers were supplying **Q1(3000 units)** quantity. As the price starts rising, the quantity supplied also starts rising .

ASSUMPTIONS IN LAW OF SUPPLY

- Income of buyers and sellers remains unchanged.
- The commodity is measurable and available in small units.
- The tastes and preferences of buyers remain unchanged.
- The cost of all factors of production does not change over a period of time.
- The time period under consideration is short.

- The technology used remains constant.
- The producer is rational.
- Natural factors remain stable.
- Expectations of producers and the government policy do not change over a period of time.